

BTEC Level 3 Applied Science • Unit 1 Chemistry • Answer sheet

Electron Configuration — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

A–E

1. 2, 8 and 8.
2. s: 2. p: 6.
3. Group 1: one outer electron. Group 7: seven outer electrons.
4. N 2,5; Mg 2,8,2; Cl 2,8,7; K 2,8,8,1.
5. O $1s^2 2s^2 2p^4$; Al $1s^2 2s^2 2p^6 3s^2 3p^1$; Ca $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$.
6. Hund: one electron in each equal-energy orbital before pairing. N $2p^3$ three single arrows; O $2p^4$ one pair plus two singles.
7. O^{2-} 2,8 / $1s^2 2s^2 2p^6$; Al^{3+} 2,8 / $1s^2 2s^2 2p^6$; Cl^- 2,8,8 / $1s^2 2s^2 2p^6 3s^2 3p^6$.
8. Calcium is 2,8,8,2. After 2,8,8 the next two electrons occupy the fourth shell / 4s, not a 10-electron third shell.
9. Mg is 2,8,2. Losing two electrons leaves 2,8, a full outer shell.
10. Mg^{2+} is 2,8 or $1s^2 2s^2 2p^6$. Full outer-shell / noble-gas arrangement is stable. [3]
11. Oxygen is 2,6. Gaining two electrons gives 2,8. [3]
12. Na s-block; F p-block; Fe d-block, with one justification. [3]
13. P is [Ne] $3s^2 3p^3$. P^{3-} is [Ne] $3s^2 3p^6$. The ion has gained three electrons to fill 3p.